Question		Answer	Marks	AO	Guidance
13	(a)	Acceleration of A is 0.75 Velocity of A is $[v_A =] -3 + 0.75t$	**B1**	**3.1b**	Correct expression for velocity from graph
		Same velocity when $t = T$ $-3 + 0.75T = 1.5 + 0.03T^2$	**M1**	**3.1a**	Equated velocities and attempts to solve.
		$T = 10$ or ~~$T = 15$~~ So $T = 10$	**A1**	**2.1**	**AG** Must see exclusion of $T = 15$ If M0A0 awarded, **SC1** can be awarded if $T = 10$ substituted into both velocity equations and are shown to be equal. Can be awarded if $v_A = 4.5$ calculated at $t = 10$ and used to calculate the time for $v_B = 4.5$ or vice versa. (Beware circular argument) "First time" equal is not established by these methods
			[3]		

Question		Answer	Marks	AO	Guidance
13	(b)	At $T = 10$ A has travelled a distance $\left\|\frac{1}{2}(-3 \times 4)\right\| + \frac{1}{2}(10-4) \times 4.5$	M1	3.4	Attempting to find at least one area under the velocity-time graph
			A1	1.1	At least one part correct allowing sign errors
		$= 6 + 13.5 = 19.5$ m	A1	1.1	soi
		At $T = 10$ B has travelled $\int_0^{10} (1.5 + 0.03t^2)\,\mathrm{d}t = 25$	M1 A1	3.1a 1.1	Attempts definite integral to calculate distance Cao
		So B has travelled 5.5 m further than A	A1FT	1.1	FT their distances (dep on both method marks)
		Alternative method for finding the distance for A $u = -3, a = 0.75, t = 4$ so $s = -6$ $u = 0, a = 0.75, t = 10 - 4$ so $s = 13.5$	M1 A1		Uses *suvat* equations for $0 < t < 4$ or $4 < t < 10$ or $0 < t < 10$ using $u = -3$, and their $a = 0.75$, Correct displacement or distance for either part
		So total distance is 19.5 m	A1		soi
		Alternative method for finding the distance for A $\int_0^4 (-3 + 0.75t)\,\mathrm{d}t = -6$ $\int_4^{10} (-3 + 0.75t)\,\mathrm{d}t = 13.5$	M1 A1		Attempt to find definite integral $0 < t < 4$ or $4 < t < 10$ or $0 < t < 10$. FT their v_A Correct displacement or distance for either part
		So total distance is 19.5 m	A1		soi BC may give total area so allow $\int_0^{10} (-3 + 0.75t)\,\mathrm{d}t = 19.5$
			[6]		